

Model Question Paper Set - 1 with effect from 2022(CBCS Scheme)

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Fourth Semester B.E Degree Examination

DISCRETE MATHEMATICAL STRUCTURES (BCS405A)

TIME: 03Hours

Max.Marks:100

Note:

1. Answer any **FIVE** full questions, choosing at least **ONE** question from each **MODULE**
2. M: Marks, L: RBT levels, C: Course outcomes.

Module – 1			M	L	C
Q.1	a	Show that the compound proposition $[(p \leftrightarrow q) \wedge (q \leftrightarrow r) \wedge (r \leftrightarrow p)] \Leftrightarrow [(p \rightarrow q) \wedge (q \rightarrow r) \wedge (r \rightarrow p)]$ for primitive statements p, q, r is logically equivalence.	6	L2	CO1
	b	Establish the validity of the following argument using the Rules of Inference: $\{p \wedge (p \rightarrow q) \wedge (s \vee r) \wedge (r \rightarrow \sim q)\} \rightarrow (s \vee t)$.	7	L2	CO1
	c	For the universe of all integers, let p(x), q(x), r(x), s(x) and t(x) denote the following open statements: p(x): $x > 0$, q(x): x is even, r(x): x is a perfect square, s(x): x is divisible by 3, t(x): x is divisible by 7. Write the following statements in symbolic form: i) At least one integer is even. ii) There exists a positive integer that is even. iii) If x is even, then x is not divisible by 3. iv) No even integer is divisible by 7. v) There exists even integer divisible by 3.	7	L1	CO1
OR					
Q.2	a	Define a tautology. Prove that, for any propositions p, q, r the compound propositions, $\{(p \rightarrow q) \wedge (q \rightarrow r)\} \rightarrow \{(p \rightarrow r)\}$ is tautology.	6	L2	CO1
	b	Prove the following using laws of logic: $p \rightarrow (q \rightarrow r) \Leftrightarrow (p \wedge q) \rightarrow r$.	7	L2	CO1
	c	Give i) direct proof ii) indirect proof iii) proof by contradiction for the following statement: “if n is an odd integer then n+9 is an even integer”.	7	L3	CO1
Module – 2					
Q.3	a	Prove that $1^2 + 3^2 + 5^2 + \dots + (2n - 1)^2 = \frac{n(2n+1)(2n-1)}{3}$ by Mathematical Induction.	6	L2	CO2
	b	Let $a_0 = 1, a_1 = 2, a_2 = 3$ and $a_n = a_{n-1} + a_{n-2} + a_{n-3}$ for $n \geq 3$. Prove that $a_n \leq 3^n \forall n \in \mathbb{Z}^+$.	7	L2	CO2
	c	Find the number of ways of arrangement of the letters of the word ‘TALLAHASSEE’ which have no adjacent A’s.	7	L2	CO2
OR					
Q.4	a	Determine the coefficient of xyz^2 in the expansion of $(2x - y - z)^4$.	6	L2	CO2
	b	In how many ways one can distribute 8 identical marbles in 4 distinct containers so that i) no container is empty ii) the fourth container has an odd number of marbles in it.	7	L2	CO2
	c	How many positive integers n can we form using the digits 3,4,4,5,5,6,7 if we want n to exceed 5,000,000?	7	L2	CO2
Module – 3					
Q.5	a	Let $f: \mathbb{R} \rightarrow \mathbb{R}$ be defined by, $f(x) = \begin{cases} 3x - 5 & \text{if } x > 0 \\ 1 - 3x & \text{if } x \leq 0 \end{cases}$. Find $f^{-1}(0), f^{-1}(1), f^{-1}(-1), f^{-1}(3), f^{-1}(6), f^{-1}([-6, 5])$ and $f^{-1}([-5, 5])$	6	L2	CO3

	b	State Pigeon hole principle. Prove that in any set of 29 persons; at least 5 persons have been born on the same day of the week.	7	L2	CO3
	c	Let $A=\{1,2,3,4,6\}$ and 'R' be a relation on 'A' defined by aRb if and only if "a is multiple of b" represent the relation 'R' as a matrix, draw the diagram and relation R.	7	L2	CO3
OR					
Q.6	a	If $f:A \rightarrow B, g:B \rightarrow C, h:C \rightarrow D$ are three functions, then Prove that $h \circ (g \circ f) = (h \circ g) \circ f$.	6	L2	CO3
	b	Show that if $n+1$ numbers are chosen from 1 to $2n$ then at least one pair add to $2n+1$.	7	L2	CO3
	c	Draw the Hasse diagram representing the positive divisors of 72.	7	L2	CO3
Module – 4					
Q.7	a	In how many ways the 26 letters of English alphabet are permuted so that none of the pattern's CAR, DOG, PUN or BYTE occurs?	6	L2	CO4
	b	Define Derangement. In how many ways can each of 10 people select a left glove and a right glove out of a total of 10 pairs of gloves so that no person selects a matching pair of gloves?	7	L2	CO4
	c	Solve the recurrence relation: $C_n = 3C_{n-1} - 2C_{n-2}$, for $n \geq 2$, given $C_1 = 5, C_2 = 3$.	7	L3	CO4
OR					
Q.8	a	In how many ways one can arrange the letters of the word CORRESPONDENTS so that there are i) exactly 2 pairs of consecutive identical letters? ii) at least 3 pairs of consecutive identical letters? iii) no pair of consecutive identical letters?	6	L2	CO4
	b	Find the rook polynomial for the chess board as shown in the figure 	7	L2	CO4
	c	Solve the recurrence relation $a_{n+2} - 3a_{n+1} + 2a_n = 0, a_0 = 1, a_1 = 6$.	7	L3	CO4
Module – 5					
Q.9	a	If H, K are subgroups of a group G , prove that $H \cap K$ is also a subgroup of G . Is $H \cup K$ a subgroup of G .	6	L2	CO5
	b	Define Klein 4 group. Verify $A = \{1, 3, 5, 7\}$ is a Klein 4 group.	7	L2	CO5
	c	State and prove Lagrange's Theorem.	7	L2	CO5
OR					
Q.10	a	Show that i) the identity of G is unique. ii) the inverse of each element of G is Unique.	6	L3	CO5
	b	Show that $(A, \cdot)$ is an abelian group where $A = \{a \in \mathbb{Q} a \neq -1\}$ and for any $a, b \in A, a \cdot b = a + b + ab$.	7	L3	CO5
	c	Let $G = S_4$, for $\alpha = \begin{pmatrix} 1 & 2 & 3 & 4 \\ 2 & 3 & 4 & 1 \end{pmatrix}$, find the subgroup $H = \langle \alpha \rangle$. Determine the left cosets of H in G .	7	L3	CO5

Model Question Paper Set - 2 with effect from 2022(CBCS Scheme)

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Fourth Semester B.E Degree Examination

DISCRETE MATHEMATICAL STRUCTURES (BCS405A)

TIME:03Hours

Max.Marks:100

Note:

1. Answer any **FIVE** full questions, choosing at least **ONE** question from each **MODULE**
2. M: Marks, L: RBT levels, C: Course outcomes.

		Module - 1	M	L	C
Q.1	a	Define tautology. Show that $[(p \vee q) \wedge (p \rightarrow r) \wedge (q \rightarrow r)] \rightarrow r$ is a tautology by constructing the truth table.	6	L1	CO1
	b	Prove the following using the laws of logic $[\neg p \wedge (\neg q \wedge r)] \vee [(q \wedge r) \vee (p \wedge r)] \Leftrightarrow r$.	7	L2	CO1
	c	For any two odd integers m and n, show that (i) m+n is even (ii) mn is odd.	7	L2	CO1
OR					
Q.2	a	Define i) an open statement ii) Quantifiers	6	L2	CO1
	b	Write the following argument in symbolic form and then establish the validity If A gets the Supervisor's position and works hard, then he will get a raise. If he gets a raise, then he will buy a car. He has not purchased a car. Therefore he did not get the Supervisor's position or he did not work hard.	7	L1	CO1
	c	For the following statements, the universe comprises all non-zero integers. Determine the truth value of each statement. a) $\exists x \exists y [xy = 1]$ b) $\exists x \forall y [xy = 1]$ c) $\forall x \exists y [xy = 1]$ d) $\exists x \exists y [(2x + y = 5) \wedge (x - 3y = -8)]$ e) $\exists x \exists y [(3x - y = 7) \wedge (2x + 4y = 3)]$	7	L2	CO1
Module – 2					
Q.3	a	Define the well ordering principle. By Mathematical Induction, Prove that $(n!) \geq 2n-1$ for all integers $n \geq 1$.	6	L2	CO2
	b	Prove that every positive integer $n \geq 24$ can be written as a sum of 5's and/or 7's.	7	L3	CO2
	c	How many positive integers n , can we form using the digits 3,4,4,5,5,6,7, if we want n to exceed 5,000,000.	7	L1	CO2
OR					
Q.4	a	By Mathematical Induction Prove that $1.3 + 2.4 + \dots + n(n+2) = \frac{n(n+1)(2n+7)}{6}$.	6	L1	CO2
	b	Find the number of permutations of the letters of the word MASSASAUGA. In how many of these all four A's are together? How many of them begin with S?	7	L2	CO2
	c	i) Obtain the Coefficient of a^5b^2 in the expansion of $(2a-3b)^7$ ii) Using the Binomial theorem find the coefficient of x^5y^2 in	7	L1	CO2

		the expansion of $(x + y)^7$.			
Module – 3					
Q.5	a	State Pigeon –hole principle. Prove that if any number from 1 to 8 are chosen then two of them will have their sum as 9.	6	L1	CO3
	b	Let $f: R \rightarrow R$ be defined by $f(x) = \begin{cases} 3x - 5, & \text{if } x > 0 \\ 1 - 3x, & \text{if } x \leq 0 \end{cases}$ find, $f^{-1}([-6,5])$ and $f^{-1}([-5,5])$.	7	L1	CO3
	c	Let $A = B = C = R$, and $f: A \rightarrow B$ and $g: B \rightarrow C$ be defined by $f(a) = 2a + 1$, $g(b) = \frac{1}{3}b, \forall a \in A, \forall b \in B$. Compute $g \circ f$ and show that $g \circ f$ is invertible. What is $(g \circ f)^{-1}$?	7	L2	CO3
OR					
Q.6	a	Let f and g be functions from R to R defined by $f(x) = ax + b$ and $g(x) = 1 - x + x^2$. If $(g \circ f)(x) = 9x^2 - 9x + 3$ determine a and b .	6	L3	CO3
	b	Draw the Hasse (POSET) diagram which represents positive divisors of 36.	7	L2	CO3
	c	Let $A = \{1,2,3,4,6\}$ and R be a relation on A defined by aRb if and only if “ a is a multiple of b ”. Write down the relation R , relation matrix $M(R)$ And draw its diagram. List out its in degree and out degree.	7	L2	CO3
Module – 4					
Q.7	a	Determine the number of positive integers n such that $1 \leq n \leq 100$ and n is not divisible by 2, 3, or 5	6	L2	CO4
	b	In how many ways can the 26 letters of the English alphabet be permuted so that none of the patterns CAR, DOG, PUN or BYTE occurs?	7	L2	CO4
	c	Solve the recurrence relation $a_n = na_{n-1}$ where $n \geq 1$ and $a_0 = 1$.	7	L2	CO4
OR					
Q.8	a	In how many ways 5 number of a's, 4 number of b's and 3 number of c's can be arranged so that all the identical letters are not in a single block?	6	L3	CO4
	b	Five teachers T_1, T_2, T_3, T_4, T_5 are to be made class teachers for five classes, C_1, C_2, C_3, C_4, C_5 , one teacher for each class. T_1 and T_2 do not wish to become the class teachers for C_1 or C_2 , T_3 and T_4 for C_4 or C_5 , and T_5 for C_3 or C_4 or C_5 . In how many ways can the teachers be assigned the work (without displeasing any teacher)	7	L2	CO4
	c	Solve the recurrence relation $F_{n+2} = F_{n+1} + F_n$ where $n \geq 0$ and $F_0 = 0, F_1 = 1$.	7	L2	CO4
Module – 5					
Q.9	a	Define group. Show that fourth roots of unity is an abelian group.	6	L2	CO5
	b	If G be a set of all non-zero real numbers and let $a * b = ab/2$ then show that $(G, *)$ is an abelian group.	7	L2	CO5
	c	Define Klein 4-group. And if $A = \{e, a, b, c\}$ then show that this is a Klein -4 group	7	L1	CO5
OR					
Q.10	a	Define Cyclic group and show that $(G, 8)$ whose multiplication table is as given below is Cyclic	6	L2	CO5

		<table><tr><td>*</td><td>a</td><td>b</td><td>c</td><td>d</td><td>e</td><td>f</td></tr><tr><td>a</td><td>a</td><td>b</td><td>c</td><td>d</td><td>e</td><td>f</td></tr><tr><td>b</td><td>b</td><td>c</td><td>d</td><td>e</td><td>f</td><td>a</td></tr><tr><td>c</td><td>c</td><td>d</td><td>e</td><td>f</td><td>a</td><td>b</td></tr><tr><td>d</td><td>d</td><td>e</td><td>f</td><td>a</td><td>b</td><td>c</td></tr><tr><td>e</td><td>e</td><td>f</td><td>a</td><td>b</td><td>c</td><td>d</td></tr><tr><td>f</td><td>f</td><td>a</td><td>b</td><td>c</td><td>d</td><td>e</td></tr></table>	*	a	b	c	d	e	f	a	a	b	c	d	e	f	b	b	c	d	e	f	a	c	c	d	e	f	a	b	d	d	e	f	a	b	c	e	e	f	a	b	c	d	f	f	a	b	c	d	e			
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b	State and prove Lagrange's theorem						7	L1	CO5																																													
c	If G be a group with subgroup H and K. If $ G = 660$ and $ K = 66$ and $K \subset H \subset G$ and find the possible value for $ H $						7	L2	CO5																																													

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